

HEALTH-KEY ULTRA — Production Datasheet

EoS Health · Model EOS-HKU-001 · Revision 1.0

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1. Product Overview

HEALTH-KEY ULTRA is a USB-C pendant health monitor that delivers clinical-grade ECG, SpO₂, blood alcohol estimation, UV index sensing, and real-time biometric analysis in a form factor worn around the neck or clipped to clothing. It connects simultaneously via USB-C (wired) and BLE 5.2 (wireless) to the EoS Health Hub mobile application.

2. Electrical Specifications

Parameter	Min	Typ	Max	Unit
Supply voltage (USB-C VBUS)	4.75	5.0	5.25	V
Battery voltage (Li-Po)	3.0	3.7	4.2	V
Battery capacity	—	210	—	mAh
Active current (all sensors on)	—	28.5	35	mA
BLE advertising current	—	4.2	—	mA
Deep sleep current	—	6.2	8	µA
USB-C charging current	—	500	1000	mA
Charging time (0 → 100%)	—	25	35	min

3. Sensor Suite

Sensor	IC	Measurement	Accuracy	Sampling Rate
ECG (2-lead)	ADS1292R	Lead I, Lead II, derived Lead III	SNR ≥ 63 dB, CMRR ≥ 100 dB	500 Hz
Pulse Oximetry	MAX30102	SpO ₂ , Heart Rate	ARMS $\leq 0.44\%$ (ISO 80601-2-61)	100 Hz
Blood Alcohol	MQ-3B	BAC estimation (breath)	$\pm 0.01\%$ BAC	1 Hz
UV Index	VEML6075	UVA, UVB, UV Index	± 1 UV Index unit	1 Hz
Accelerometer	LSM6DSO	3-axis motion, step count	± 0.1 g	104 Hz
Temperature	MCP9808	Skin surface temperature	$\pm 0.25^{\circ}\text{C}$	1 Hz

4. Wireless Specifications

Parameter	Value
BLE version	5.2
PHY	1M / 2M (auto-negotiated)
Frequency band	2.402–2.480 GHz
TX power	–20 to +8 dBm (configurable)
Typical range (open air)	100 m
Antenna	Molex 2137600100 chip antenna
Security	LE Secure Connections (LESC), AES-128-CCM
BLE services	EoS Health GATT (UUID: EOS1), Battery (0x180F), Device Info (0x180A)

5. Mechanical Specifications

Parameter	Value
Dimensions	62 × 22 × 9 mm
Weight	18 g (without lanyard)
Housing material	Anodized aluminum 6061-T6
Connector	USB-C 2.0 (data + charging)
IP rating	IP68 (1 m, 30 min)
Operating temperature	−10°C to +50°C
Storage temperature	−20°C to +60°C
Drop resistance	1.5 m onto concrete (MIL-STD-810H Method 516.8)
Lanyard attachment	Stainless steel swivel clip

6. Processor and Memory

Component	Part	Specification
MCU	Nordic nRF52840	Cortex-M4F, 64 MHz, 1 MB Flash, 256 KB RAM
NVM	W25Q32JV	4 MB SPI Flash (firmware + data buffer)
Secure element	ATECC608B	Ed25519 key storage, OTA verification
USB bridge	CP2102N	USB-C to UART, CDC ACM

7. Power Budget

Mode	Current	Duration	Energy
Active (all sensors, BLE connected)	28.5 mA	5 min/hr	2.375 mAh/hr
Idle (BLE advertising, IMU only)	4.2 mA	50 min/hr	3.500 mAh/hr
Deep sleep	6.2 μ A	5 min/hr	0.001 mAh/hr
Weighted average	~28.8 mA	—	—
Battery life	—	~7.3 days	—

8. Firmware

Component	Version	Description
EoS Firmware	1.0.0	Production release
MCUboot bootloader	1.10.0	Dual-bank OTA, Ed25519 verification
BLE stack	S140 v7.3.0	Nordic SoftDevice
nRF5 SDK	17.1.0	Peripheral drivers
TFLite Micro	2.14.0	On-device AFib inference

OTA update: Signed firmware packages delivered via BLE (SUIT manifest, Ed25519). Rollback automatic on boot failure. Minimum battery 20% required.

9. Regulatory Compliance

Standard	Status
FCC Part 15 (BLE)	Certification pending
CE RED 2014/53/EU	Certification pending
FDA 510(k)	Pre-submission Q4 2026
RoHS 3 (2015/863/EU)	Compliant
REACH SVHC	Compliant
IEC 60601-1 (electrical safety)	Testing Q4 2026
ISO 80601-2-61 (SpO ₂)	Validated (ARMS 0.44%)
ANSI/AAMI EC11 (ECG)	Validated (SNR 63.5 dB)

10. BOM Summary (Production, 1k units)

Category	Key Components	Unit Cost
MCU + memory	nRF52840, W25Q32JV, ATECC608B	\$6.20
Sensors	ADS1292R, MAX30102, MQ-3B, VEML6075, LSM6DSO, MCP9808	\$9.80
Power	BQ25185 PMIC, MAX17048 fuel gauge, 210 mAh Li-Po	\$4.10
RF	Molex chip antenna, matching network	\$0.85
Connectivity	CP2102N USB bridge, USB-C connector	\$1.20
Passives + PCB	Resistors, caps, inductors, 4-layer PCB	\$3.40
Housing + assembly	Aluminum enclosure, lanyard, packaging	\$8.50
Total BOM		\$34.05
Target retail		\$249

11. Ordering Information

Model	Description	SKU
EOS-HKU-001-BLK	HEALTH-KEY ULTRA, Black Anodized	EOS-HKU-001-BLK
EOS-HKU-001-SLV	HEALTH-KEY ULTRA, Silver Anodized	EOS-HKU-001-SLV
EOS-HKU-001-DEV	Developer Kit (device + J-Link EDU Mini + breakout board)	EOS-HKU-001-DEV